

The Model Confidence Set

Part IV ■ Multiple Testing and Simulation-Based Inference

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The Problem

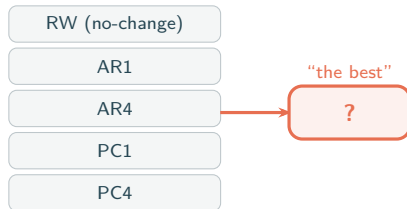
The model proliferation problem

Econometricians routinely face a large set of competing models $\mathcal{M}^0 = \{m_1, m_2, \dots, m_{m_0}\}$.

The question *Which is the best?* is often unanswerable with a single, unambiguous choice:

- Many models perform similarly in finite samples
- Selecting one ignores the *uncertainty* in that choice
- The “winner” can flip with the loss function

Example. 12-month-ahead US CPI inflation: AR models with $p = 1, 2, 4, 8, 12$ lags, Phillips curve variants, and a naive no-change benchmark: 10 specifications, often statistically indistinguishable.



⋮ 10 specifications

AIC/BIC: picks exactly one

**Selecting a single model ignores the uncertainty in that selection.
The MCS replaces a point choice with a confidence set.**

Limits of existing tools

AIC / BIC

Select a single model by penalised likelihood.

Silent about the uncertainty in that selection. No coverage guarantee.

Pairwise tests

Compare models two at a time (e.g., Diebold–Mariano).

Do not control the familywise error rate across the family of comparisons.

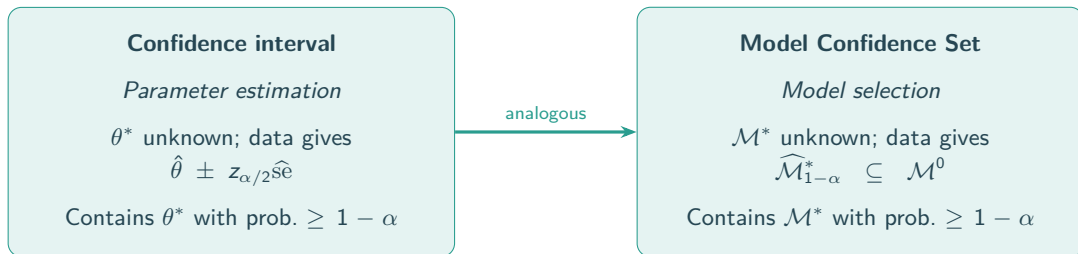
Reality Check / SPA

Test whether a *pre-specified benchmark* is beaten.

Requires a benchmark. Yields one rejection decision, not a set of best models.

What we want. A procedure that **(i)** reduces $\mathcal{M}^0$ to a smaller set containing the best model(s) with probability $\geq 1 - \alpha$, **(ii)** is honest about uncertainty, returning a large set when data are uninformative.

The confidence-interval analogy



Large $\hat{se}$
 $\Rightarrow$ wide CI

same logic

Uninformative data
 $\Rightarrow$ large MCS

A large MCS is not a failure: it is an honest answer.
The data cannot always identify a unique best model, and the MCS says so.

Theory

The superior set $\mathcal{M}^*$ and the MCS

What are we trying to find? Given a set of competing models $\mathcal{M}^0$ and a loss function, the **superior set** $\mathcal{M}^*$ contains every model that is not strictly worse than any other in expected loss. There may be more than one: ties are allowed.

$\mathcal{M}^*$ depends on unknown expected losses, so we cannot observe it directly. The MCS builds $\widehat{\mathcal{M}}_{1-\alpha}^*$, a data-based estimate of $\mathcal{M}^*$ that contains the true superior set with probability at least $1 - \alpha$.

Three things to know about $\mathcal{M}^*$.

- It is specific to $\mathcal{M}^0$: the MCS says nothing about models outside the set you started with.
- A unique best model is the special case $|\mathcal{M}^*| = 1$.
- When data are uninformative, $\widehat{\mathcal{M}}_{1-\alpha}^*$ can equal all of $\mathcal{M}^0$: the MCS is honest about that.

$\mathcal{M}^*$ is the target. The MCS is the data-based confidence set that contains it.

A large MCS is honest, not a failure: the data cannot always identify a unique winner.

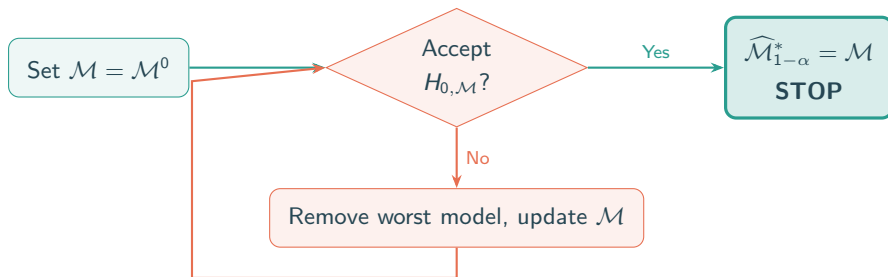
The MCS algorithm

The procedure (Hansen, Lunde & Nason, 2011)

Step 0. Set $\mathcal{M} = \mathcal{M}^0$.

Step 1. Test $H_{0,\mathcal{M}}$: all remaining models are equally good, using an equivalence test at level α .

Step 2. If $H_{0,\mathcal{M}}$ *accepted*: set $\widehat{\mathcal{M}}_{1-\alpha}^* = \mathcal{M}$ and **stop**. If $H_{0,\mathcal{M}}$ *rejected*: remove the worst model; return to Step 1.



How the MCS controls FWER

The key question. How can the MCS apply many sequential tests and still control FWER at α ?

The answer: sequential halting. A false elimination requires rejecting $H_{0,\mathcal{M}}$ when $\mathcal{M}$ contains only superior models. Condition (c) of the algorithm ensures that superior models are never eliminated while inferior ones remain, so $\mathcal{M}$ can only contain superior models at the step where the procedure halts. At that step $H_{0,\mathcal{M}}$ is true, the test is at level α , and the probability of a false rejection is therefore bounded by α .

Connection to Part I. This is the same logic as the closure principle: the sequential structure forces any false elimination to pass through a single α -level test under the true null. One valid test, one bound.

FWER is controlled because a false elimination can only occur at the halting step, where the null is true and the test is at level α .

Coherency: matching the test to the elimination rule

A rejection tells us *someone* is inferior, not *who*. The elimination rule must be matched to the test so that the model removed is the one most likely driving the rejection.

“Another analogy is that it is easier to conclude that a murder has taken place, than it is to determine who committed the murder.”

— Hansen, Lunde & Nason (2011, fn. 1)

Mismatched (test, rule) pairs can eliminate a superior model and break the coverage guarantee. The `MCSprocedure()` function in R always uses a coherent pair automatically.

**A rejection identifies that *someone* is inferior.
Coherency ensures the elimination rule names the right one.**

The elimination rule produces a sequence of nested sets. The **MCS p -value** for each eliminated model is the running maximum of the test p -values up to that step.

The key result (HLN 2011)

Model i is in $\widehat{\mathcal{M}}_{1-\alpha}^*$ **if and only if** its MCS p -value is at least α .

The running-maximum construction enforces monotonicity: MCS p -values never decrease as models are eliminated. The last surviving model always receives MCS p -value = 1 by convention.

Read the MCS p -value like a regular p -value:

$\hat{p}_i \geq \alpha$ means model i survives; $\hat{p}_i < \alpha$ means it is eliminated.

MCS p -values: worked example

	Step	Test p -value	MCS p -value	Status
$\alpha = 0.05$	$e\mathcal{M}_1$	0.010	0.010	eliminated
	$e\mathcal{M}_2$	0.040	0.040	eliminated
	$e\mathcal{M}_3$	0.020	0.040	eliminated
	$e\mathcal{M}_4$	0.030	0.040	eliminated
	$e\mathcal{M}_5$	0.070	0.070	in SSM
	$e\mathcal{M}_6$	0.040	0.070	in SSM
	$e\mathcal{M}_7$	0.110	0.110	in SSM
	$e\mathcal{M}_8$	0.250	0.250	in SSM
	$e\mathcal{M}_9$	1.000	1.000	in SSM

MCS p -value = running maximum of test p -values $\Rightarrow$ monotone by construction

Bootstrap and Connections

Why bootstrap?

Two test statistics. $T_{\max}$ compares each model against the *average* loss across all competitors. T_R takes the largest absolute *pairwise* gap. $T_{\max}$ is preferred when m is large as it avoids estimating the full covariance matrix; T_R can be sharper when one model strongly dominates another. Both are reported in the output; the bottom p-value tracks whichever is chosen (`statistic = "Tmax"` in our demo).

Why bootstrap? The null distribution of either statistic depends on an unknown correlation matrix across model losses, which cannot be derived analytically. Resampling blocks of the observed loss differentials $\{d_{ij,t}\}$ estimates this distribution without ever computing the matrix. The bootstrap p -value counts how often the simulated statistic exceeds the observed one.

Why blocks? Loss differentials are serially correlated. Resampling whole blocks preserves the dependence structure, giving an asymptotically valid approximation to the null.

The stationary bootstrap handles serial dependence and unknown correlation. Both $T_{\max}$ and T_R are reported; the bottom p-value tracks the chosen statistic.

How the null is imposed: centring the bootstrap

The null hypothesis. $H_{0,\mathcal{M}}$ says all models in $\mathcal{M}$ have equal expected loss: $\mu_{ij} = 0$ for all $i, j \in \mathcal{M}$.

The problem. The observed loss differentials $\bar{d}_{ij}$ are generally nonzero even when $H_{0,\mathcal{M}}$ is true, simply because of sampling variability. If we resample directly from $\{d_{ij,t}\}$, the bootstrap distribution reflects the observed data, not the null.

The fix: recentre before resampling. In each bootstrap replication, subtract the observed mean before computing the test statistic:

$$t_{i\cdot,b}^* = \frac{\bar{d}_{i\cdot}^* - \bar{d}_{i\cdot}}{\widehat{\text{var}}^{1/2}(\bar{d}_{i\cdot}^*)}.$$

The subtraction of $\bar{d}_{i\cdot}$ imposes $\mu_{ij} = 0$ on the bootstrap world. The simulated statistic then has asymptotically the same distribution as $t_{i\cdot}$ would under the null, giving an asymptotically valid reference distribution for the test.

**Recentring the bootstrap statistics imposes the null.
Without it, the bootstrap p -value would be asymptotically invalid.**

MCS vs Reality Check and SPA

Reality Check

White, 2000

Tests: is the *pre-specified benchmark* outperformed?

Requires a benchmark.

Returns: one binary decision.

SPA Test

Hansen, 2005

Improved power over RC via better centring.

Still requires a benchmark.

Returns: one binary decision.

MCS

HLN, 2011

No benchmark needed.

Returns: a *set* of superior models.

Naturally handles $|\mathcal{M}^*| > 1$.

Connection to Parts I–III. The MCS sequential structure mimics a step-down procedure: it controls FWER through sequential halting without any Bonferroni penalty. MCS p -values are monotone by construction, in the same spirit as step-down adjusted p -values.

**The MCS needs no benchmark and returns a set rather than a binary decision.
FWER is controlled for free through sequential halting.**

Application

Multiple comparisons in forecast evaluation

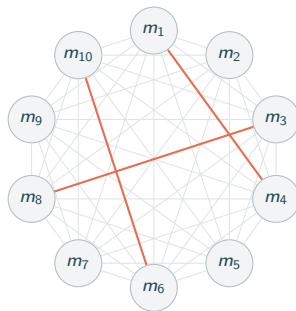
With m forecasting models, naive pairwise evaluation generates $\binom{m}{2}$ hypothesis tests simultaneously.

For $m = 10$: $\binom{10}{2} = \mathbf{45}$ pairwise comparisons.

Testing each at $\alpha = 0.05$, the probability of at least one false rejection under the global null is $1 - (0.95)^{45} \approx \mathbf{0.90}$.

The “winner” may be the luckiest model, not the best.

The MCS solution. Replace 45 uncontrolled pairwise tests with one coherent sequential procedure that controls FWER and returns an honest set of best models.



$\binom{10}{2} = 45$ pairwise tests
FWER uncontrolled

The horse race: data

Data (FRED, monthly 1959–2000):

Forecast target (stored at origin t):

$$\pi_t^{(12)} = 100 \log \frac{\text{CPI}_{t+12}}{\text{CPI}_t}$$

12-month-ahead cumulative CPI inflation.

Phillips curve predictor:

$$\text{ugap}_t = \text{UNRATE}_t - \text{NAIRU}_t$$

CBO natural rate of unemployment interpolated to monthly frequency. No look-ahead bias.

Evaluation: direct $h = 12$ OLS, rolling window $W = 120$ months.

Loss function: absolute forecast error $\ell_{i,t} = |\pi_t^{(12)} - \hat{\pi}_{i,t}^{(12)}|$.

The horse race: 10 competing models

No-change benchmark (1 model):

$$\hat{\pi}_t^{(12)} = \pi_{t-12}^{(12)}$$

Uses last year's inflation rate as the forecast. No estimation.

AR models (5 models):

$$\pi_t^{(12)} = a + \sum_{j=1}^p b_j \pi_{t-j+1} + e_t$$

Direct OLS, $p \in \{1, 2, 4, 8, 12\}$ lags of monthly inflation.

Phillips curve models (4 models):

$$\pi_t^{(12)} = a + \sum_{j=1}^p b_j \pi_{t-j+1} + c \text{ugap}_t + e_t$$

AR augmented with the unemployment gap, $p \in \{1, 2, 4, 8\}$.

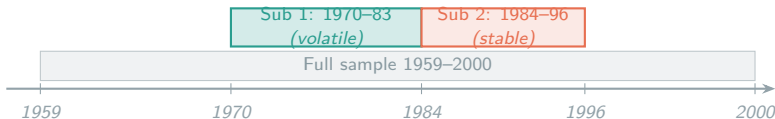
Group	Models
RW	no-change
AR	AR1, AR2, AR4, AR8, AR12
PC	PC1, PC2, PC4, PC8
Total	10 models

AR models ask: does past inflation predict future inflation?

PC models ask: does labour market slack add information?

The horse race: evaluation design

Direct $h = 12$ OLS forecasting with a **rolling** window of $W = 120$ months.



Loss: absolute forecast error

$$\ell_{i,t} = \left| \pi_t^{(12)} - \hat{\pi}_{i,t}^{(12)} \right|$$

Yields an $(n \times 10)$ loss matrix.

Two regimes. Sub 1: high, volatile inflation.

Sub 2: great-moderation inflation, harder to beat a simple benchmark.

The two subsamples partition the *evaluation* period, not the estimation period. At each forecast origin the rolling window looks back $W = 120$ months, so early Sub-sample 2 forecasts are estimated on Sub-sample 1 data. The window transitions gradually across the 1984 break.

MCS as the disciplined referee

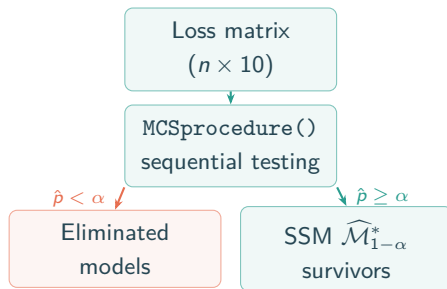
The loss matrix feeds directly into `MCSprocedure()`.

What MCS does instead of 45 pairwise tests:

1. Tests $H_{0,\mathcal{M}}$: all remaining models equally good
2. If rejected: removes the worst model ($e_{\mathcal{M}}$)
3. Repeats until the null is accepted

FWER control: sequential halting bounds the probability of eliminating any superior model at α , across all steps simultaneously, without Bonferroni.

MCS p -values: running maxima of step-level p -values, monotone by construction, equivalent to step-down Holm adjusted p -values.



Large SSM: data uninformative
Small SSM: clear winners identified

The MCS replaces 45 uncontrolled pairwise tests with one coherent procedure that controls FWER and returns an honest set of best models.

Demo 4: MCS on US CPI inflation forecasts

`demo_mcs_cpi.R`

- 10 competing models: random walk, AR(1–12), Phillips curve variants.
- Two subsamples: volatile 1970s (Sub 1) and great moderation (Sub 2).
- Loss function: absolute forecast error at $h = 12$, $B = 5000$ bootstrap replications.
- Watch the MCS size: large in Sub 1, singleton in Sub 2.

Reading the output (1): the columns

Two test statistics, two sets of rankings.

	Rank_M	v_M	MCS_M	Rank_R	v_R	MCS_R	Loss
	← based on $T_{\max}$: model vs. average →			← based on T_R : best pairwise gap →			
RW	9	0.558	0.962	8	0.776	0.991	2.444
AR8	10	0.788	0.864	10	2.708	0.072	2.416
PC4	1	-0.814	1.000	1	-0.092	1.000	2.267
	(... 7 other models...)						

$T_{\max}$ columns (M).

Rank_M: rank by standardised loss relative to the group average $\bar{d}_i$. Rank 10 = most excess loss.

v_M: the standardised excess loss t_i .

MCS_M: MCS p -value under $T_{\max}$.

T_R columns (R).

Rank_R: rank by the largest pairwise loss gap $\sup_j t_{ij}$. Rank 10 = most dominated by some other model.

v_R: the standardised pairwise gap.

MCS_R: MCS p -value under T_R .

Loss: mean absolute forecast error.

Reading the output (2): Sub-sample 1

Sub-sample 1 (1970–1983): all 10 models survive.

	Rank_M	v_M	MCS_M	Rank_R	v_R	MCS_R	Loss
RW	9	0.558	0.962	8	0.776	0.991	2.444
AR1	8	0.484	0.982	9	0.837	0.985	2.408
AR2	4	-0.233	1.000	4	0.360	1.000	2.326
AR4	6	-0.141	1.000	5	0.388	1.000	2.330
AR8	10	0.788	0.864	10	2.708	0.072	2.416
AR12	7	0.310	0.998	7	0.680	1.000	2.376
PC1	5	-0.183	1.000	6	0.443	1.000	2.313
PC2	2	-0.673	1.000	2	0.092	1.000	2.275
PC4	1	-0.814	1.000	1	-0.092	1.000	2.267
PC8	3	-0.527	1.000	3	0.116	1.000	2.274

p-value: 0.864 = bootstrap p under $T_{\max}$ at the halting step = smallest MCS_M. Null accepted: all models survive.

AR8 (rank 10): most excess loss and most dominated pairwise; would exit first if the null were rejected.

PC4 (rank 1): best raw loss. Despite a 0.18 gap in mean loss, no pair is separable: p-value 0.864.

A large MCS is the honest answer when the bootstrap cannot separate the models.